# Case Analysis – Avoiding Misleading Proportions

*Activity 6: Questions and Answers*

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## Question 1: Truncated Y-Axis in Sales Growth

A retail company presents the following monthly sales (₹ lakhs). The marketing team creates a bar chart with the Y-axis starting at 90 instead of 0, visually suggesting June sales are almost twice January's.

| Month | Sales (₹ lakhs) |
|---|---|
| January | 95 |
| February | 97 |
| March | 99 |
| April | 101 |
| May | 103 |
| June | 105 |

**Task 1:** ***Why is the chart misleading?***

The actual increase from January (95) to June (105) is only about 10.5%. By starting the Y-axis at 90 instead of 0, the visible bar heights are stretched: January's bar covers only 5 units above the baseline while June's covers 15 units — a visual ratio of 3:1 even though the real data ratio is only about 1.1:1. The truncated axis exaggerates the apparent growth and makes a modest, steady increase look dramatic.

**Task 2:** ***Which visualization principle has been violated?***

This violates the principle of a zero-based (true) baseline, closely tied to Tufte's "Lie Factor" concept — the idea that the size of a visual effect shown in a graphic should be proportional to the size of the effect in the underlying data. Bar charts encode magnitude through bar length starting at zero; truncating the axis breaks that proportional encoding and misrepresents relative magnitude.

**Task 3:** ***Recommend an improved visualization method.***

- Use a bar chart with the Y-axis starting at 0, so bar heights are proportional to actual sales values.
- If the goal is to highlight a small month-to-month change, use a line chart (starting at 0) with data labels showing exact figures, or a separate chart/table showing percentage growth rather than distorting the primary chart.
- Clearly annotate the axis scale and add gridlines so viewers can read absolute values accurately.

**Task 4:** ***How could this misleading chart influence business decisions?***

Management could overestimate the strength of sales momentum and make decisions based on false confidence — for example, over-ordering inventory, increasing production capacity, approving larger marketing budgets, or promising unrealistic growth to investors and stakeholders. When actual performance later reveals only modest growth, this can lead to wasted resources, missed targets, and loss of credibility with investors or the board.

## Question 2: Pie Chart with Incorrect Proportions

A university reports student enrollment across four departments. The designer accidentally creates a pie chart with slices at approximately 50%, 30%, 15%, and 15% instead of the correct proportions.

| Department | Students |
|---|---|
| Computer Science | 420 |
| Electronics | 280 |
| Mechanical | 200 |
| Civil | 100 |

**Task 1:** ***Calculate the correct percentage for each department.***

Total students = 420 + 280 + 200 + 100 = 1000.

| Department | Students | Correct % |
|---|---|---|
| Computer Science | 420 | 42% |
| Electronics | 280 | 28% |
| Mechanical | 200 | 20% |
| Civil | 100 | 10% |

**Task 2:** ***Explain why the displayed chart is misleading.***

The displayed slices (50%, 30%, 15%, 15%) do not match the correct values (42%, 28%, 20%, 10%) and, notably, they add up to 110% instead of 100% — an internal inconsistency that itself proves the chart is inaccurate. Computer Science and Electronics appear inflated while Mechanical and Civil appear compressed or mismatched, distorting the true relative sizes of departments and giving viewers an incorrect mental model of enrollment distribution.

**Task 3:** ***Suggest an alternative chart that would improve comparison accuracy.***

- A horizontal or vertical bar chart, sorted by enrollment, makes it much easier to compare exact magnitudes than pie-slice angles.
- If proportion-of-whole is essential, use a 100% stacked bar chart with data labels showing exact percentages.
- Always validate that segment values are computed from and sum correctly to the underlying data before rendering.

**Task 4:** ***How could such errors affect institutional planning?***

Inaccurate enrollment proportions can distort decisions about faculty hiring, classroom and lab allocation, budget distribution, and infrastructure investment. For example, if Computer Science appears artificially larger and Civil appears artificially smaller than reality, the university might over-invest in CS resources while under-resourcing Civil Engineering, leading to overcrowded classes in one department and underutilized capacity in another.

### Question 3: Unequal Bar Widths in Revenue Comparison

A software company compares annual revenue across four products. Instead of equal-width bars, the designer uses increasing bar widths, making Product D appear nearly three times larger than Product A.

| Product | Revenue (₹ Crores) |
|---|---|
| A | 60 |
| B | 75 |
| C | 90 |
| D | 105 |

**Task 1:** ***Identify the graphical error.***

The error is varying bar width (and consequently bar area) along with bar height to represent a single-variable comparison. Revenue differences among the four products are modest (a maximum ratio of 105:60, or about 1.75:1), but because both width and height increase together, the visual area difference becomes far more extreme than the underlying data justifies.

**Task 2:** ***Explain how unequal bar widths distort perception.***

Viewers perceive the two-dimensional area of a bar, not just its height. When width increases along with height, the perceived area grows roughly with the square of the linear increase. A bar that is, say, 1.75 times taller and 1.75 times wider appears roughly 3 times larger in area — which explains why Product D looks nearly three times the size of Product A even though the actual revenue difference is far smaller.

**Task 3:** ***Redesign the visualization by specifying the correct design principles.***

- Use bars of equal, fixed width for all categories; only the height (or length) should vary with the data value.
- Start the value axis at zero so height is directly proportional to revenue.
- Keep consistent spacing between bars and use a single scale/gridlines across the chart.
- Add data labels with exact revenue figures to remove any ambiguity.

**Task 4:** ***Discuss the possible consequences if investors rely on this chart.***

Investors could dramatically overestimate the performance gap between products, leading them to misjudge which product lines are truly driving growth. This might result in misallocated investment toward Product D, reduced confidence or funding for Product A despite it being reasonably competitive, and ultimately poor capital allocation decisions based on a visually exaggerated — rather than factual — picture of the business.

### Question 4: 3D Column Chart Distortion

An energy company reports renewable energy generation using a 3D column chart with perspective effects. Due to the viewing angle, the Hydro bar appears almost 40% larger than Wind, although the actual difference is only 10%.

| Source | Energy (GWh) |
|---|---|
| Solar | 450 |
| Wind | 500 |
| Hydro | 550 |
| Biomass | 520 |

**Task 1:** ***Explain why 3D charts can create misleading proportions.***

3D charts introduce perspective, depth, and rotation effects that distort how bar height and depth are perceived. Bars positioned closer to the viewer (in the foreground) can appear larger due to foreshortening, while bars further back may be partly occluded or visually compressed. This depth-based distortion has nothing to do with the actual data values, yet it changes how large or small a column appears to the eye.

**Task 2:** ***Compare the perceived difference with the actual difference.***

Actual difference: Hydro (550) vs Wind (500) is a difference of 50 GWh, or exactly 10%. Perceived difference: due to the 3D perspective effect, Hydro appears roughly 40% larger than Wind. This means the visual distortion overstates the real difference by roughly four times, giving a false impression of Hydro's dominance in the renewable mix.

**Task 3:** ***Recommend a better visualization.***

- Use a standard flat 2D column or bar chart with a zero-based axis so heights are directly proportional to GWh values.
- Order bars logically (e.g., by source type or by magnitude) and use consistent, non-perspective bar widths.
- Add precise data labels above each bar to eliminate reliance on visual estimation altogether.

**Task 4:** ***Discuss the importance of avoiding visual distortion in sustainability reporting.***

Sustainability reports are used by regulators, investors, and the public to assess a company's environmental performance and credibility. Distorted visuals — even if unintentional — can be perceived as an attempt to mislead or "greenwash," damaging trust and inviting regulatory scrutiny. Accurate, undistorted charts are essential for transparent reporting, informed policy-making, and maintaining stakeholder confidence in the company's environmental claims.

## Question 5: Bubble Chart with Incorrect Scaling

A healthcare organization compares disease cases across five districts. The analyst scales bubble radius directly to the number of cases rather than scaling the bubble area, making District E appear more than ten times larger than District A.

| District | Cases |
|---|---|
| A | 150 |
| B | 250 |
| C | 350 |
| D | 450 |
| E | 550 |

**Task 1:** ***Explain why scaling the radius instead of the area is misleading.***

Viewers judge the size of a bubble by its overall visible area, not its radius. The area of a circle grows with the square of the radius (Area = $\pi \times r^2$). If radius is scaled directly and linearly to the data value, then the visible area scales with the square of that value. District E (550 cases) has about 3.7 times the cases of District A (150 cases), but if radius scales linearly, the area ratio becomes roughly $(550/150)^2 \approx 13.4$ — which is consistent with District E appearing "more than ten times larger" even though the true difference is under 4 times.

**Task 2:** ***Describe the correct method for proportional bubble sizing.***

The bubble's area, not its radius, should be directly proportional to the data value. This means the radius must be scaled using the square root of the value: $r = \sqrt{(\text{value} \div \pi \times k)}$, where k is a constant scaling factor. This way, when the underlying value doubles, the visible area (and hence perceived size) also doubles — rather than quadrupling.

**Task 3:** ***Suggest an alternative visualization suitable for this dataset.***

- A simple sorted bar chart is often clearer than a bubble chart for comparing a single value (cases) across a handful of districts.
- If geographic context matters, use a choropleth (color-shaded) map or a correctly area-scaled proportional symbol map.
- Whichever chart is used, include exact case counts as data labels to avoid reliance on visual size estimation.

**Task 4:** ***Discuss the potential impact on public health resource allocation.***

If District E's disease burden is visually exaggerated relative to District A, health authorities may over-allocate medical staff, beds, and funding to District E while under-resourcing other districts whose actual case counts are also significant but appear disproportionately small. This can lead to inequitable distribution of healthcare resources, delayed response in under-visualized districts, and ultimately poorer health outcomes driven by a data visualization error rather than genuine need.